

Run your research on Nautilus, from your AI assistant

Connect your laptop to the National Research Platform, give your assistant the **nrp** tools, then move a Slurm job to Kubernetes by asking in plain language.

National GPUs and CPUs

No UCR recharge and no allocation proposal to write.

Plain language

Your assistant plans, runs, watches and cleans up for you.

Nothing hidden

Every plan shows what it creates and the kubectl it equals.

Asks first

Running, publishing and deleting each need a plan you approve.

1 Sign in to Nautilus

2 min

Go to **nrp.ai**, click **Login**, choose **University of California, Riverside**, and accept the policy. Your first sign-in is what activates your namespace invite.

✓ You see your name on **nrp.ai**; your namespace admin sees you.

2 Install nrp-mcp

2 min

MAC / LINUX

```
$ curl -fsSL https://raw.githubusercontent.com/UCR-Research-Computing/nrp-mcp/main/scripts/install.sh | sh
$ echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.zshrc # macOS; Linux: ~/.bashrc
$ export PATH="$HOME/.local/bin:$PATH"
```

WINDOWS

Download the `windows_amd64.zip` from the **Releases** page, then paste the PowerShell lines from the web version (KB031, QR on the back). They put `nrp-mcp.exe` on your PATH.

✓ `nrp-mcp version` prints **0.7.0 or newer**.

3 Connect your laptop

8 min

Download your config at **nrp.ai/config** (it lands in Downloads), then run:

```
$ nrp-mcp setup
```

It installs kubectl and the sign-in plugin into your user folder (no admin rights, checksums verified), puts the config in place, then opens a normal browser tab for the UCR sign-in. It asks before changing anything.

✓ It ends with **"This computer is ready for Nautilus."**

4 Install Hermes Agent and add your key RECOMMENDED

8 min

MAC / LINUX

```
$ curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

WINDOWS

```
PS> iex (irm https://hermes-agent.nousresearch.com/install.ps1)
```

Open a new terminal. Store your Gemini API key and pick the model:

```
$ hermes config set GEMINI_API_KEY your-key
$ hermes config set model.provider gemini
$ hermes config set model.default gemini-3.8-flash
```

✓ `hermes chat -q "hello"` answers.

5 Give Hermes the nrp tools

2 min

```
$ hermes mcp add nrp --command "$HOME/.local/bin/nrp-mcp" --args serve
```

Windows: `--command "$env:LOCALAPPDATA\Programs\nrp-mcp\bin\nrp-mcp.exe"`. Answer **Y** to enable all 9 tools; if it cannot connect, answer **N** to "save anyway" and fix the path. Then start `hermes` and ask:

"Where do I stand on Nautilus?"

✓ It answers with your user, namespace, what is running and your GPU quota.

Move a Slurm array job to Nautilus in two sentences

The exercise

Get `examples/slurm-array-bootstrap` from the project (QR below, Code, Download ZIP): an R bootstrap of a trimmed mean and a normal Slurm script with `--array=0-9`. In that folder, ask Hermes:

1. “Plan the Slurm script in this folder and show me what each line becomes.”
2. “Run it.”
3. “How did it go?”
4. “Clean up everything I made.”

SLURM LINE	BECOMES ON NAUTILUS
<code>--cpus-per-task=1</code> <code>--mem=2G</code>	CPU and memory request; limits equal requests
<code>--time=00:20:00</code>	a time limit on the Job (whole hours)
<code>--array=0-9</code>	an Indexed Job: 10 tasks, each with its own <code>\$SLURM_ARRAY_TASK_ID</code>
<code>module load R</code>	a container image with R in it

Ten tasks, ten seeds, ten different answers, in about a minute. The same script still runs on Slurm.

What to say

YOU SAY	NRP DOES
“Where do I stand?”	who you are, what runs, quotas, warnings
“Run train.py on a GPU”	plans it, waits for your yes, then runs
“Run this 50 times, seeds 1 to 50”	a sweep (Indexed Job)
“Why did it fail?”	logs plus a plain diagnosis and the fix
“Give me a Jupyter notebook”	a private session on Nautilus
“Copy results to my laptop”	downloads from your volume
“Clean up”	lists what you made, deletes after a 2nd yes

Rules of the road

- **P1 data only.** Public, non-sensitive data. No student records, PII, clinical or controlled data.
- **No idle GPUs.** Ask for a GPU only when the code uses one; idle GPUs get accounts suspended.
- **Nothing public without asking RC.** Publishing a web app needs you to type its URL back.
- **Clean up** when you finish. Jobs left running hold resources others need.
- **Keys stay private.** Never paste a key into a chat, a repo, a shared doc or a screenshot.

If you get stuck

“no Nautilus context”	download the config at <code>nrp.ai/config</code> , rerun <code>nrp-mcp setup</code>
“forbidden” / not in namespace	sign in at <code>nrp.ai</code> once more; ask your namespace admin
<code>command not found</code>	run the PATH lines in step 2, or use <code>~/local/bin/nrp-mcp</code>
“Not ready yet: kubectl”	an older kubectl (Docker Desktop) is first on PATH; ask RC
Hermes has no nrp tools	<code>hermes mcp test nrp</code> ; if it fails, remove and redo step 5
Laptop will not cooperate	use the NRP JupyterHub (only needs the sign-in): jupyterhub-west.nrp-nautilus.io

Prefer another assistant?

Gemini CLI (Node.js 20+): `npm i -g @google/gemini-cli`, set `GEMINI_API_KEY`, then `gemini mcp add nrp ~/.local/bin/nrp-mcp serve`

OpenCode: set `GOOGLE_GENERATIVE_AI_API_KEY`, model `google/gemini-3.8-flash`, add nrp under `mcp` in `opencode.json`. Claude Code and others: see KB031.



nrp-mcp

Install, Releases (Windows zip) and the exercise in `examples/` github.com/UCR-Research-Computing/nrp-mcp



Your NRP config

Sign in first, then Get Config; it lands in Downloads nrp.ai/config



This sheet, online

Copyable commands, Windows steps, other assistants; links to the full Nautilus guide RC Knowledge Base, KB031